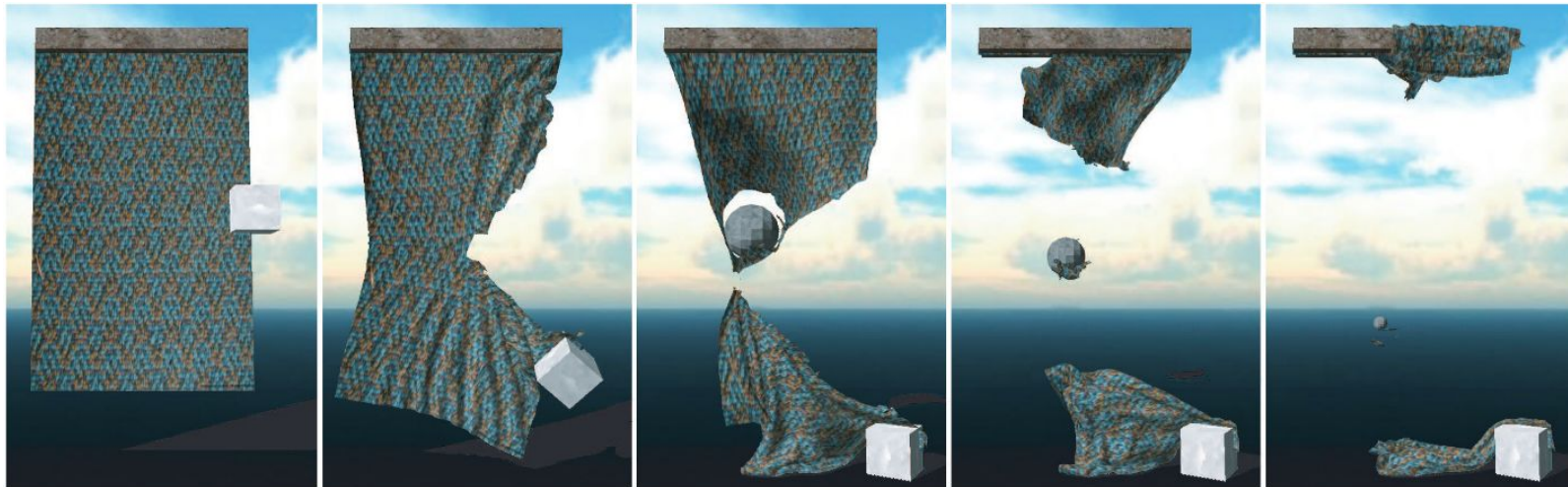
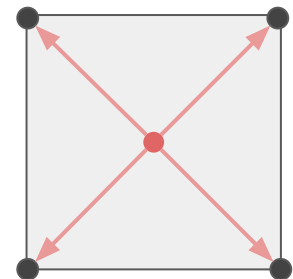
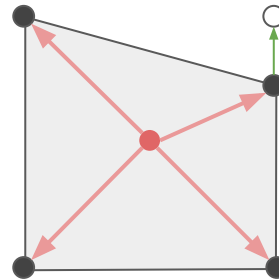
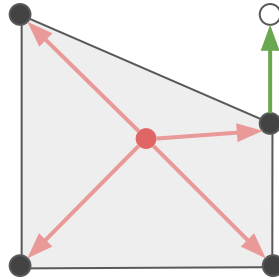
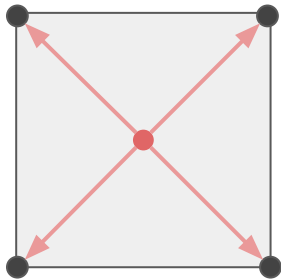
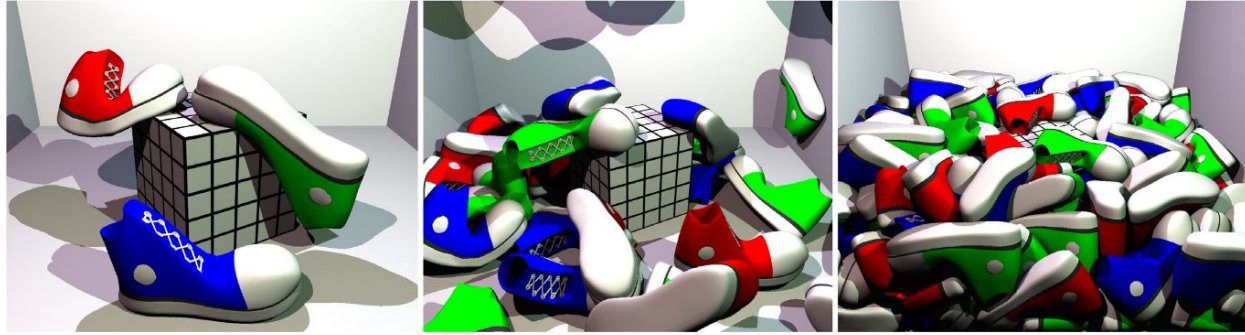


Position Based Dynamics (2006)

Matthias Müller, Bruno Heidelberger, Marcus Hennix, John Ratcliff



Meshless Deformations Based on Shape Matching (2005)



stiffness : α [0,1] 0 $\rightarrow$ no stiff 1 $\rightarrow$ rigid body

Position Based Dynamics (2006)

Particle/Vertex $i \in [1, \dots, N]$

position : $\mathbf{x}_i$

velocity : $\mathbf{v}_i$

mass : m_i

Constraints $j \in [1, \dots, M]$

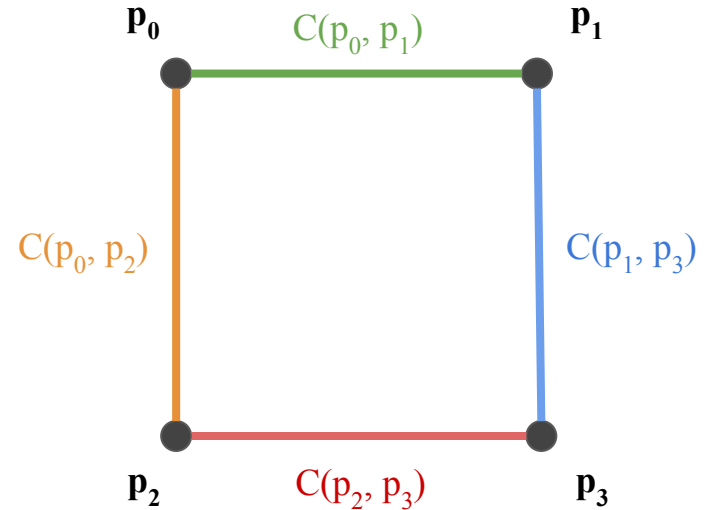
cardinality : n

indices : $\{i_1, \dots, i_{n_j}\}, i_k \in [1, \dots, N]$

fonction : $C_j(\mathbf{x}_{i_1}, \dots, \mathbf{x}_{i_{n_j}}) \rightarrow \mathbb{R}$

stiffness : $k \in [0, 1]$

condition : *equality* ($C = 0$) or *inequality* ($C \geq 0$)



PBD Solver

While(True)

forall vertices i do $v_i \leftarrow v_i + \Delta t w_i f_{\text{ext}}$

dampVelocities($v_1, \dots, v_N$)

forall vertices i do $p_i \leftarrow x_i + \Delta t v_i$

forall vertices i do generateCollisionConstraints($x_i \rightarrow p_i$)

loop solverIterations times

projectConstraints($C_1, \dots, C_{M+M_{\text{coll}}}, p_1, \dots, p_N$)

endloop

forall vertices i

$v_i \leftarrow (p_i - x_i) / \Delta t$

$x_i \leftarrow p_i$

endfor

velocityUpdate($v_1, \dots, v_N$)

endloop

Compute external interactions

Damp velocities

Compute position

Check collision and create constraints

Project particles to valid positions

Compute new velocity

Set projected position

Fix broken velocities

PBD Solver

```
While(True)
  forall vertices i do  $v_i \leftarrow v_i + \Delta t w_i f_{\text{ext}}$ 

  dampVelocities( $v_1, \dots, v_N$ )

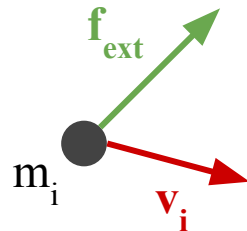
  forall vertices i do  $p_i \leftarrow x_i + \Delta t v_i$ 

  forall vertices i do generateCollisionConstraints( $x_i \rightarrow p_i$ )

  loop solverIterations times
    projectConstraints( $C_1, \dots, C_{M+M_{\text{coll}}}, p_1, \dots, p_N$ )
  endloop

  forall vertices i
     $v_i \leftarrow (p_i - x_i) / \Delta t$ 
     $x_i \leftarrow p_i$ 
  endfor
  velocityUpdate( $v_1, \dots, v_N$ )

endloop
```



$$w_i = 1 / m_i \quad a_i = f_{\text{ext}} w_i$$

PBD Solver

```
While(True)
  forall vertices i do  $\mathbf{v}_i \leftarrow \mathbf{v}_i + \Delta t \mathbf{w}_i \mathbf{f}_{\text{ext}}$ 

  dampVelocities( $\mathbf{v}_1, \dots, \mathbf{v}_N$ )

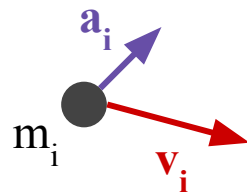
  forall vertices i do  $\mathbf{p}_i \leftarrow \mathbf{x}_i + \Delta t \mathbf{v}_i$ 

  forall vertices i do generateCollisionConstraints( $\mathbf{x}_i \rightarrow \mathbf{p}_i$ )

  loop solverIterations times
    projectConstraints( $\mathbf{C}_1, \dots, \mathbf{C}_{M+M_{\text{coll}}}, \mathbf{p}_1, \dots, \mathbf{p}_N$ )
  endloop

  forall vertices i
     $\mathbf{v}_i \leftarrow (\mathbf{p}_i - \mathbf{x}_i) / \Delta t$ 
     $\mathbf{x}_i \leftarrow \mathbf{p}_i$ 
  endfor
  velocityUpdate( $\mathbf{v}_1, \dots, \mathbf{v}_N$ )

endloop
```



$$\mathbf{w}_i = 1 / m_i \quad \mathbf{a}_i = \mathbf{f}_{\text{ext}} \mathbf{w}_i$$

$$\mathbf{v}_i = \mathbf{v}_i + \Delta t \mathbf{a}_i$$

PBD Solver

While(True)

forall vertices i do $\mathbf{v}_i \leftarrow \mathbf{v}_i + \Delta t \mathbf{w}_i \mathbf{f}_{\text{ext}}$

dampVelocities($\mathbf{v}_1, \dots, \mathbf{v}_N$)

forall vertices i do $\mathbf{p}_i \leftarrow \mathbf{x}_i + \Delta t \mathbf{v}_i$

forall vertices i do generateCollisionConstraints($\mathbf{x}_i \rightarrow \mathbf{p}_i$)

loop solverIterations times

projectConstraints($\mathbf{C}_1, \dots, \mathbf{C}_{M+M_{\text{coll}}}, \mathbf{p}_1, \dots, \mathbf{p}_N$)

endloop

forall vertices i

$\mathbf{v}_i \leftarrow (\mathbf{p}_i - \mathbf{x}_i) / \Delta t$

$\mathbf{x}_i \leftarrow \mathbf{p}_i$

endfor

velocityUpdate($\mathbf{v}_1, \dots, \mathbf{v}_N$)

endloop



$$\mathbf{w}_i = 1 / m_i \quad \mathbf{a}_i = \mathbf{f}_{\text{ext}} \mathbf{w}_i$$

$$\mathbf{v}_i = \mathbf{v}_i + \Delta t \mathbf{a}_i$$

PBD Solver

While(True)

forall vertices i do $\mathbf{v}_i \leftarrow \mathbf{v}_i + \Delta t \mathbf{w}_i \mathbf{f}_{\text{ext}}$

dampVelocities($\mathbf{v}_1, \dots, \mathbf{v}_N$)

forall vertices i do $\mathbf{p}_i \leftarrow \mathbf{x}_i + \Delta t \mathbf{v}_i$

forall vertices i do generateCollisionConstraints($\mathbf{x}_i \rightarrow \mathbf{p}_i$)

loop solverIterations times

projectConstraints($\mathbf{C}_1, \dots, \mathbf{C}_{M+M_{\text{coll}}}, \mathbf{p}_1, \dots, \mathbf{p}_N$)

endloop

forall vertices i

$\mathbf{v}_i \leftarrow (\mathbf{p}_i - \mathbf{x}_i) / \Delta t$

$\mathbf{x}_i \leftarrow \mathbf{p}_i$

endfor

velocityUpdate($\mathbf{v}_1, \dots, \mathbf{v}_N$)

endloop



$$\mathbf{v}_i = \mathbf{v}_i + ???$$

$$\mathbf{v}_i = \mathbf{v}_i - k_{\text{damp}} \mathbf{v}_i \quad k_{\text{damp}} \in [0, 1]$$

PBD Solver

While(True)

forall vertices i do $\mathbf{v}_i \leftarrow \mathbf{v}_i + \Delta t \mathbf{w}_i \mathbf{f}_{\text{ext}}$

dampVelocities($\mathbf{v}_1, \dots, \mathbf{v}_N$)

forall vertices i do $\mathbf{p}_i \leftarrow \mathbf{x}_i + \Delta t \mathbf{v}_i$

forall vertices i do generateCollisionConstraints($\mathbf{x}_i \rightarrow \mathbf{p}_i$)

loop solverIterations times

projectConstraints($\mathbf{C}_1, \dots, \mathbf{C}_{M+M_{\text{coll}}}, \mathbf{p}_1, \dots, \mathbf{p}_N$)

endloop

forall vertices i

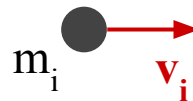
$\mathbf{v}_i \leftarrow (\mathbf{p}_i - \mathbf{x}_i) / \Delta t$

$\mathbf{x}_i \leftarrow \mathbf{p}_i$

endfor

velocityUpdate($\mathbf{v}_1, \dots, \mathbf{v}_N$)

endloop



$$\mathbf{v}_i = \mathbf{v}_i + ???$$

$$\mathbf{v}_i = \mathbf{v}_i - k_{\text{damp}} \mathbf{v}_i \quad k_{\text{damp}} \in [0,1]$$

PBD Solver

While(True)

forall vertices i do $v_i \leftarrow v_i + \Delta t w_i f_{\text{ext}}$

dampVelocities($v_1, \dots, v_N$)

forall vertices i do $p_i \leftarrow x_i + \Delta t v_i$

forall vertices i do generateCollisionConstraints($x_i \rightarrow p_i$)

loop solverIterations times

projectConstraints($C_1, \dots, C_{M+M_{\text{coll}}}, p_1, \dots, p_N$)

endloop

forall vertices i

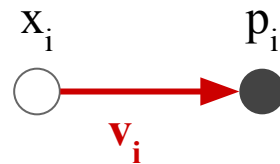
$v_i \leftarrow (p_i - x_i) / \Delta t$

$x_i \leftarrow p_i$

endfor

velocityUpdate($v_1, \dots, v_N$)

endloop



PBD Solver

While(True)

forall vertices i do $v_i \leftarrow v_i + \Delta t w_i f_{\text{ext}}$

dampVelocities($v_1, \dots, v_N$)

forall vertices i do $p_i \leftarrow x_i + \Delta t v_i$

forall vertices i do generateCollisionConstraints($x_i \rightarrow p_i$)

loop solverIterations times

projectConstraints($C_1, \dots, C_{M+M_{\text{coll}}}, p_1, \dots, p_N$)

endloop

forall vertices i

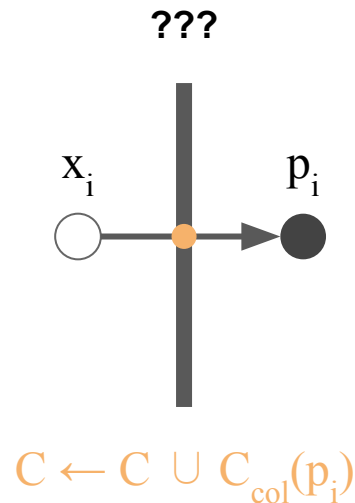
$v_i \leftarrow (p_i - x_i) / \Delta t$

$x_i \leftarrow p_i$

endfor

velocityUpdate($v_1, \dots, v_N$)

endloop



PBD Solver

While(True)

forall vertices i do $v_i \leftarrow v_i + \Delta t w_i f_{\text{ext}}$

dampVelocities($v_1, \dots, v_N$)

forall vertices i do $p_i \leftarrow x_i + \Delta t v_i$

forall vertices i do generateCollisionConstraints($x_i \rightarrow p_i$)

loop solverIterations times

projectConstraints($C_1, \dots, C_{M+M_{\text{coll}}}, p_1, \dots, p_N$)

endloop

forall vertices i

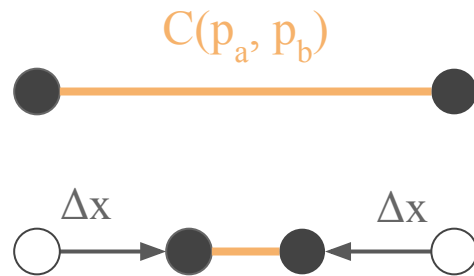
$v_i \leftarrow (p_i - x_i) / \Delta t$

$x_i \leftarrow p_i$

endfor

velocityUpdate($v_1, \dots, v_N$)

endloop



PBD Solver

While(True)

forall vertices i do $v_i \leftarrow v_i + \Delta t w_i f_{\text{ext}}$

dampVelocities($v_1, \dots, v_N$)

forall vertices i do $p_i \leftarrow x_i + \Delta t v_i$

forall vertices i do generateCollisionConstraints($x_i \rightarrow p_i$)

loop solverIterations times

projectConstraints($C_1, \dots, C_{M+M_{\text{coll}}}, p_1, \dots, p_N$)

endloop

forall vertices i

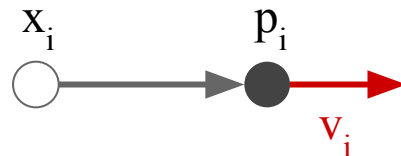
$v_i \leftarrow (p_i - x_i) / \Delta t$

$x_i \leftarrow p_i$

endfor

velocityUpdate($v_1, \dots, v_N$) ← frictions, collisions and stuff

endloop



PBD Solver

While(True)

forall vertices i do $v_i \leftarrow v_i + \Delta t w_i f_{\text{ext}}$

dampVelocities($v_1, \dots, v_N$)

forall vertices i do $p_i \leftarrow x_i + \Delta t v_i$

forall vertices i do generateCollisionConstraints($x_i \rightarrow p_i$)

loop solverIterations times

projectConstraints($C_1, \dots, C_{M+M_{\text{coll}}}, p_1, \dots, p_N$)

endloop

forall vertices i

$v_i \leftarrow (p_i - x_i) / \Delta t$

$x_i \leftarrow p_i$

endfor

velocityUpdate($v_1, \dots, v_N$)

endloop

Compute external interactions

Damp velocities

Compute position

Check collision and create constraints

Project particles to valid positions

Compute new velocity

Set projected position

Fix broken velocities

Constraints resolution

$$C(\mathbf{p} + \Delta \mathbf{p}) = 0 \quad \Delta \mathbf{p} = ???$$

Taylor order 1 :

$$f(x) \approx f(a) + f'(a)(x-a)$$

$$C(x) \approx C(\mathbf{p}) + \nabla C(\mathbf{p})(x-\mathbf{p})$$

$$C(\mathbf{p} + \Delta \mathbf{p}) \approx C(\mathbf{p}) + \nabla C(\mathbf{p}) \cdot \Delta \mathbf{p} = 0$$

$$\Delta \mathbf{p} = \lambda \nabla C(\mathbf{p})$$

$$C(\mathbf{p}) + \nabla C(\mathbf{p}) \cdot (\lambda \nabla C(\mathbf{p})) = 0$$

$$C(\mathbf{p}) + \lambda \|\nabla C(\mathbf{p})\|^2 = 0$$

$$\lambda = - \frac{C(\mathbf{p})}{\|\nabla C(\mathbf{p})\|^2}$$

Uniform masses

$$\Delta p_i = -s \nabla_{p_i} C(p_1, \dots, p_n)$$

$$s = \frac{C(p_1, \dots, p_n)}{\sum_j \|\nabla_{p_j} C(p_1, \dots, p_n)\|^2}$$

Different masses

$$\Delta p_i = -s w_i \nabla_{p_i} C(p_1, \dots, p_n)$$

$$s = \frac{C(p_1, \dots, p_n)}{\sum_j w_j \|\nabla_{p_j} C(p_1, \dots, p_n)\|^2}$$

Distance Constraint

Distance Constraint

$$C(p_1, p_2) = \| p_1 - p_2 \| - d$$



$$C'(p_1, p_2) = \nabla C(p_1, p_2) = ???$$

$$C(p_1, p_2) = [(p_1 - p_2)^T \cdot (p_1 - p_2)]^{0.5} - d$$

$$\nabla C(p_1, p_2) = \frac{1}{2(\|p_1 - p_2\|^2)^{0.5}} 2(p_1 - p_2)$$

$$(a^T \cdot a)' = 2a$$

$$(f \circ g(x))' = f' \circ g(x) g'(x)$$

$$(x^n)' = n x^{n-1}$$

$$\nabla C(p_1, p_2) = \frac{p_1 - p_2}{\|p_1 - p_2\|}$$

Distance Constraint

$$\Delta p_i = -s w_i \nabla_{p_i} C(p_1, \dots, p_n) \quad s = \frac{C(p_1, \dots, p_n)}{\sum_j w_j \|\nabla_{p_j} C(p_1, \dots, p_n)\|^2}$$

Distance constraint

$$C(p_1, p_2) = \|p_1 - p_2\| - d$$

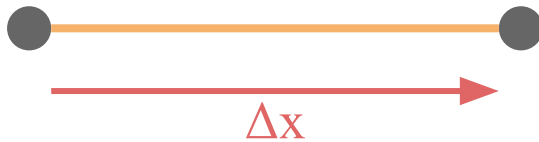
$$\nabla C(p_1, p_2) = \frac{p_1 - p_2}{\|p_1 - p_2\|}$$



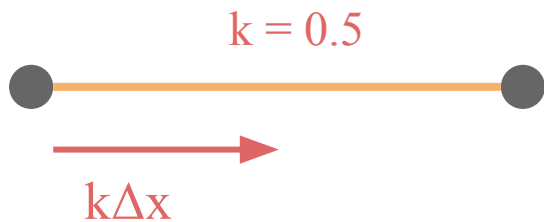
$$\Delta p_1 = -s w_1 \frac{p_1 - p_2}{\|p_1 - p_2\|}$$

$$s = \frac{\|p_1 - p_2\| - d}{w_1 + w_2}$$

Stiffness $k \in [0, 1]$



Step 1



Step 2



...

Step N ?



$$k_{\text{fixed}} = 1 - (1 - k)^{1/ns}$$

La suite sur le papier :D